WS Ray: Extending Human Perception with a Teleoperated Underwater Drone for Subaquatic Environments

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***Abstract*— Coral reef environments are ecologically and culturally valuable, highly sensitive ecosystem that play a vital role in preserving marine biodiversity. However, routine in-water inspection is challenging for both expert and non-expert users. This paper proposes WS Ray: an underwater drone for immersive reef inspection reducing the need for frequent dives. WS Ray is a prototype engineered from first principles. It is a six-propeller platform with five degrees of freedom (DOF) maneuverability and a servo-actuated, pitch-controllable camera mount that delivers an omnidirectional viewing experience while the drone executes 5-DOF motions. The tethered drone streams live video using WebRTC, onboard control is integrated with ROS2. A whale-shark-inspired hydrodynamic hull was chosen to reduce drag, and drone motion is regulated by a robust sliding-mode control (SMC). The system centers on a fixed surface station that connects multiple tethered drones and provides internet access via user-friendly operator interfaces: a game-like mobile app that streams live video that offers joystick-based teleoperation, and an immersive Virtual Reality (VR) app with head-tracked pitch/yaw control supplemented by an auxiliary handheld controller for linear motion. The prototype was tested in a 2 m depth pool environment over a local network for initial validation. The teleoperation modes provided stable control, real-time visual feedback, and an intuitive immersive experience. These results demonstrate WS Ray as a viable underwater drone for remote coral inspection and public engagement.**

I. INTRODUCTION

Shallow-sea coral environments are used by both environmentalists and tourists for purposes ranging from routine scientific monitoring to recreation and aesthetic appreciation [1]. Traditional broad-scale survey methods (e.g., manta tows, timed swims) are costly and pose risks to observers. Recreational and scientific diving introduces physiological and equipment-related hazards, such as decompression sickness, barotrauma, and hypothermia, that restrict participation and long-term monitoring programs [2]. As a result, observational practice increasingly relies on remotely operated robotic platforms: visual data alone can document the percent cover of benthic categories, fish abundance and size distributions, and the presence of disease or bleaching [3], thus reducing the dependence on in-water observers. However, robotic approaches still face practical limitations, including the need for on-site deployment and high equipment cost, which constrain survey frequency and accessibility.

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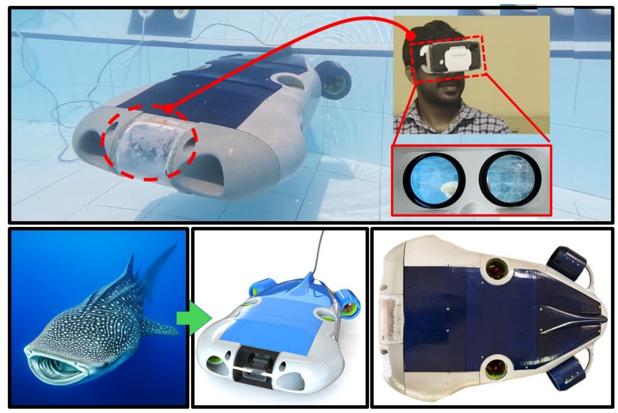


Fig. 1. WS Ray: whale-shark–inspired underwater drone streaming live camera feed to VR for teleoperation over the internet.

A networked teleoperation concept is proposed to address these challenges, featuring a fixed seafloor station that hosts multiple tethered drones. The system provides remote, internet accessible operation and real-time video feedback, enabling routine, repeatable underwater surveys and scientific monitoring. The proposed teleoperated system preserves these inspection capabilities while reducing cost and risk and simultaneously expands accessibility: scientists can conduct regular surveys remotely, and tourists or users with mobility limitations can explore reef sites through immersive interfaces. Importantly, both monitoring and recreational observation can be performed remotely from anywhere in the world, improving scalability and public engagement.

This paper presents WS Ray (see Fig. 1), a tethered underwater drone developed for use within the proposed system (see Fig. 2). The drone provides 5-DOF maneuverability and integrates a pitch-controllable, servo-actuated camera that together with coordinated drone motion that delivers an effectively omnidirectional viewing capability. The main contributions of this work are summarized as follows:

- Introduces the WS Ray underwater drone, whale-shark-inspired hydrodynamic geometry with ABS 3D-printed manufacturing, and water resistance treatment;
- Experimental and simulation-based estimation of drone properties for dynamic modeling (mass, added mass, inertia, damping);
- Two mobile apps: a game-style mobile controller (joy-

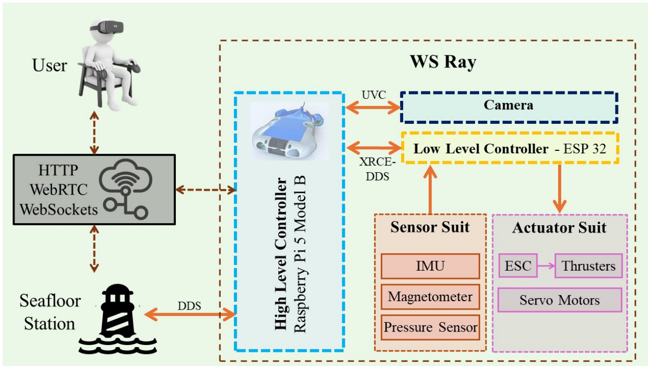


Fig. 2. System Overview: The drone's high-level controller (HLC) is tethered via Ethernet, providing a network connection for teleoperation. A dedicated DDS link to the seafloor station enables remote monitoring of the drone's status.

stick + live video) and an immersive VR app with head-tracked pitch/yaw control, plus a novel mouse-form analog-auxiliary controller for linear motion;

- Prototype tests in a 2 m depth pool environment over a local network demonstrating stable teleoperation and real-time immersive feedback.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the mechanical design of the drone that includes the whale-shark-inspired body shape. Section IV formulates the nonlinear dynamics and presents the SMC approach for drone motion. Section V describes the architecture of the embedded system. Section VI details the human–machine interface (HMI). Finally, the experimental results are discussed in Section VII and conclude the paper.

II. Related Work

Coral observation has shifted from diver led surveys to underwater robots that collect repeatable imagery and environmental data, improving safety and temporal coverage [4][5]. Reviews describe optical methods (video, photogrammetry) and pipelines in which AI accelerates benthic classification and change detection, placing robots alongside other core instruments [5][6]. Alongside conventional ROVs and AUVs, bio-inspired designs such as the soft robotic fish SoFi enable low disturbance, close range observation around reefs [7]. Fielded reef drones illustrate the range of capabilities: RangerBot, a compact vision guided AUV, supports mapping and targeted pest control on the Great Barrier Reef [8], while CUREE pairs multi camera and hydrophones with deep learning onboard for 3D habitat reconstruction and ecosystem studies [9]. Internet accessible telepresence extends exploration beyond expert operators; Teleportal enables public, browser based control of tethered reef robots with low latency video [10], and KAUST's touch sensitive "robotic avatar diver" demonstrates immersive teleoperation for deep reef science [11]. Collectively, the landscape spans affordable ROVs, autonomous survey AUVs, bio-inspired observers, and internet telepresence, forming a broad toolkit for coral observation.

Despite progress, many systems remain limited to trained users due to the cost of the equipment, mission logistics, and the need to transport and deploy drone for each survey [4][5]. Specialist teams, vessel time, and site specific preparation constrain survey frequency and reach, especially in resource limited regions [4][6]. Platforms such as RangerBot and CUREE are typically fielded by expert operators for targeted missions rather than continuous public access [8][9]. Outside Teleportal's fixed site service, most observations still require on site presence, limiting inclusivity for non divers, schools, and mobility limited users [10][11].

Reducing dependence on travel, vessel time, and favorable weather is critical for frequent, inclusive reef monitoring. Fixed-site, internet-accessible teleoperation demonstrates that users need not be on site; combined with modern optical pipelines, remote operation broadens participation, reduces per-survey cost, and improves temporal coverage, thereby motivating a networked teleoperation architecture and the development of a purpose-built underwater drone to operate from fixed seafloor stations and reliably interface with internet-accessible HMIs.

III. Structural and Hydrodynamic Design

A. Structural Layout

The structural layout was designed to achieve several key objectives, including stable handling in shallow, wave-affected water; low disturbance around coral; and manageable cost. The drone therefore adopts a six thruster arrangement (see Fig. 3) that delivers 5 DOF maneuvering—surge, sway, heave, and yaw with roll stabilized hydrostatically. Dedicated pitch actuation was omitted to avoid added mass, drag, and power; instead, a servo tilted front camera provides the visual cue of pitch independent of the drone body. This preserves the operator's perception of 6 DOF motion while keeping the propulsion system simple.

The design prioritizes a dominant forward drive axis while retaining deliberate lateral capability; because of that, a key hydrodynamic design decision was to minimize wake interference between inward-facing thrusters. Following Liu et al. [12], a spacing of more than two propeller diameters ($\geq 2D$) is maintained between adjacent inward facing units to ensure unobstructed inflow. With D = 75 mm, the naive center-to-center spacing is 2D = 150 mm. Design considerations that led to the front and rear thruster pairs are canted by θ = 30° relative to the longitudinal axis, so a purely lateral 2D spacing underestimates the required clearance because the inflow axis intersects potential obstructions downstream of the propeller disk. The effective axial separation projects into the lateral plane as lateral width $(W_{\text{lateral}}) \geq 2D/\sin\theta$. With D = 75 mm and = 30°, this gives $W_{\text{lateral}} \geq 300$ mm. Additional taking into account the mounting hardware, the wiring space, and the safety margin, the width of the chassis was set to 360 mm. The geometric ratios reported in the literature (length:width = 1.7) [13] were used as additional design references when finalizing the drone dimensions (see Fig. 3).

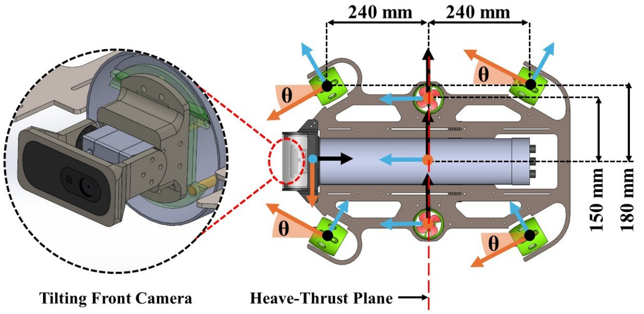


Fig. 3. WS Ray thruster layout with key dimensions and tilting front camera. Six-thruster configuration with the front and rear pairs canted 30° relative to the forward direction (θ) to provide surge–sway authority; the middle pair lies on the body mid-plane for heave.

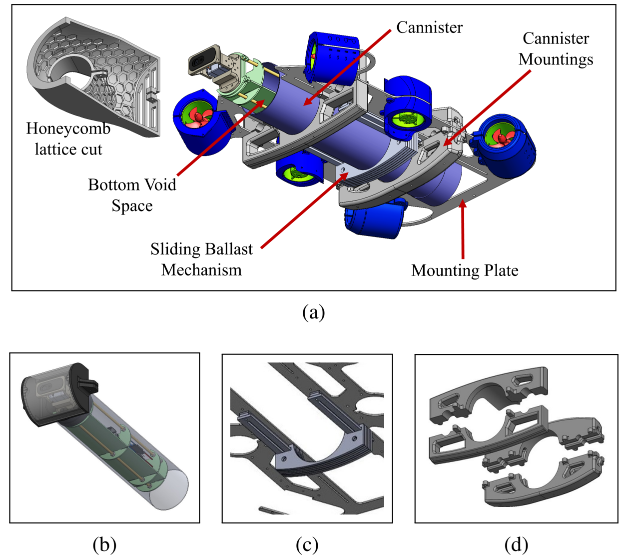


Fig. 4. Structural layout and hydrostatic features: (a) main assemblies and mountings, including the sliding ballast, canister, and mounting plate; (b) void space beneath the canister; (c) sliding ballast mechanism for longitudinal trim; (d) canister-mounting brackets, with the upper brackets having more volume than the lower pair.

The mechanical structure comprises a cylindrical canister, a rigid mounting plate that carries six propellers and their mounting brackets, and a custom front cover to accommodate the camera and viewing window (see Fig. 4). Symmetry was a primary design objective, so a cylindrical canister was chosen to minimize the asymmetric mass distribution and simplify the internal layout. The canister body is constructed from PVC, with a custom ABS 3D printed front cover. To provide a transparent forward view for the camera, a thermo-formed acrylic plate is integrated into the front cover. Electrical components and the camera are housed inside the canister; the tether and propeller wiring pass through underwater rated wire to wire connectors (penetrators) mounted on the rear end-cap, which is the only access point to the interior.

B. Buoyancy and Mass Distribution

Mass and buoyancy considerations drove the internal packaging choices: the selected canister volume is $\sim$6000 cm^3 and the combined mass of the components of the embedded system located inside the canister is approximately 1 kg. To achieve neutral buoyancy the drone requires a configuration with greater mass within the available volume. To ensure a rigid, low-vibration propulsion mounting, the thrusters and electronics are fixed to a 3 mm mild steel mounting plate, which reduces vibration transmission and provides a stiff base for the thruster assembly. The thick mild steel mounting plate alone did not fully address the mass shortfall.

Beyond neutral buoyancy, hydrostatic stability requires the center of gravity (COG) to lie below the center of buoyancy (COB). In addition, the COG, COB, and the two vertical thrusters used for heave should lie in the same plane (heave-thruster plane), which makes internal packaging and mass placement a nontrivial design task. To move the COG downward, the remaining internal volume of the canister (see Fig. 4b) was filled with 1.2 kg of steel ball bearings (diameter 7 mm). These added masses both aided neutral buoyancy and shifted the COG below the COB. The layout of the embedded system inside the canister can displace the COG from the heave-thruster plane. As development proceeds iteratively, the insertion or removal of components can alter mass distribution and create further imbalances, WS Ray employs a sliding ballast mechanism (see Fig. 4c). This system allows incremental ballast to be added, helps keep the COG lower, and by translating the ballast along a rail, it brings the COG into the heave-thruster balancing plane.

C. Whale-shark-inspired Hydrodynamic Hull Design

Hull geometry dominates drag and wake for small underwater drones, so three biologically inspired forms; manta ray, whale shark, and tuna were first compared in body only CFD (see Fig. 5). The whale shark inspired shape produced the lowest steady drag and the smallest turbulent wake cross-sectional area, and was selected for refinement. Optimization targeted two high impact geometric variables: the shoulder radius (governing fore-body curvature and separation) and the rear half taper angle (controlling base pressure recovery and wake size). The optimized whale shark fusiform reduced steady drag from 60.70 N (baseline) to 35.62 N ($\sim$41% reduction) under equivalent conditions and contracted the high turbulence wake area from $\sim$0.09 m^2 to $\sim$0.076 m^2 (15.6% reduction). These improvements were consistent in additional runs at higher cruise speeds.

The whale shark inspired hull is split and manufactured part wise for assembly (see Fig. 6) and access, with an internal hexagonal (honeycomb) lattice cut (see Fig. 4a) into body panels to increase stiffness while saving weight, and key load paths reinforced locally. Low-infill 3D printed canister mounting brackets (see Fig. 4d) lift the COB slightly above the mid-plane, keeping the COB collinear with the heave-thruster plane and thereby compensating for the loss of axial symmetry introduced by the whale-shark shape. The hull parts are printed in ABS for toughness, chemical resistance, and relative non-toxicity in shallow marine environments at reasonable cost. Wet surfaces and seams are coated with

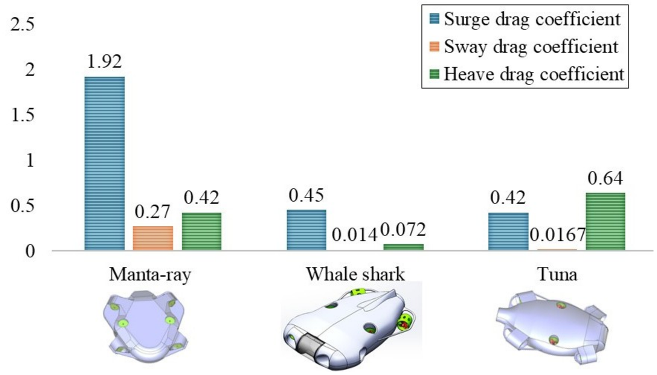


Fig. 5. Comparative hydrodynamic performance of bio-inspired planforms (body-only CFD). Drag coefficients in surge, sway, and heave for manta ray, whale shark, and tuna

Fig. 6. Whale-shark-inspired modular hull in exploded view, showing the split body panels and removable fairings

a marine grade epoxy resin that provides durable water resistance and can be reapplied or sanded for repairs.

IV. Nonlinear Marine Vehicle Dynamics and Robust Control Design

Shallow sea operation represents a highly nonlinear and disturbance-rich environment, where traditional linear controllers such as PID offer limited robustness and adaptability [14]. To address these challenges, while still meeting the requirements of low computational cost and fast response, SMC was adopted as the primary control framework [15].

A. SMC Synthesis and Control Implementation

The controller is formulated within the general SMC architecture introduced by Slotine et al. (1983) [16].

The 6-DOF nonlinear dynamics of the underwater drone are expressed as:

$$M\dot{v} + C(v)v + D(v)v + g(\eta) + \tau_e = \tau, \tag{1}$$

$$\dot{\eta} = J(\eta_2)v, \tag{2}$$

where M is the inertia matrix including added mass, $C(v)$ is the Coriolis and centripetal matrix, $D(v)$ represents hydrodynamic damping, $g(\eta)$ captures restoring forces due to buoyancy and gravity, τ_e denotes external environmental disturbances, and τ is the control input vector. The vector v represents body-fixed velocities, η the earth-fixed position and orientation, and $J(\eta_2)$ the transformation matrix between body and earth-fixed frames.

The sliding surface is defined to regulate the tracking error e as:

$$\dot{\eta}_r = \dot{\eta}_d + \lambda e \rightarrow s = \dot{\eta} - \dot{\eta}_r \tag{3}$$

where η_d is the desired trajectory, η_r is the reference trajectory, λ is a positive gain vector, and s is the sliding surface vector. Maintaining the system on this surface ensures that $\eta(t)$ converges asymptotically to $\eta_d(t)$.

To derive the control law, a Lyapunov-like candidate function is introduced:

$$V(s,t) = \tfrac{1}{2}s^T M_0(\eta)s, \tag{4}$$

where $M_0(\eta)$ is the nominal inertia matrix. By applying the Lyapunov stability criterion on the sliding surface, the control input τ is designed to satisfy $\dot{V} < -\varepsilon\|s\|$ with $\varepsilon > 0$, yielding

$$\tau = \tau_{eq} + \tau_{dis}, \tag{5}$$

with

$$\tau_{eq} = M_0(\eta)\ddot{\eta}_r + C_0(v,\eta)\dot{\eta}_r + D_0(v,\eta)\dot{\eta} - g_0(\eta), \tag{6}$$

$$\tau_{dis} = -K\,\mathrm{sgn}(s), \tag{7}$$

where τ_{eq} represents the equivalent control ensuring nominal trajectory tracking, and τ_{dis} is a discontinuous term that compensates for uncertainties and external disturbances. Here K denotes the disturbance rejection gain.

Finally, the control vector τ is mapped to individual thruster forces u via the affine thruster configuration matrix B:

$$\tau = Bu, \tag{8}$$

where u is the vector of thruster forces commanded. This allocation distributes the control effort across the six thrusters to achieve robust motion in all degrees of freedom.

In the teleoperation context the control objective is given by velocity commands. At each control step the drone's current position is taken as the origin of a local earth-fixed frame so that the user velocity inputs map directly to $\dot{\eta}_d$. The sliding variable s is formed from the tracking error according to (3). Depending on the error magnitude, the SMC law computes the control vector as the sum of the equivalent control and the disturbance-rejection term, as specified in (6)–(7). This control vector is converted into individual thruster commands via the thrust-configuration relationship (8); the resulting forces are then allocated to the corresponding onboard thrusters and applied at the next actuator update.

B. System Identification and Hydrodynamic Coefficients

For SMC, several drone parameters that were required for dynamic modeling were estimated using a combination of simulation and controlled physical experiments. CFD-based identification procedures were carried out in ANSYS Fluent

TABLE I

WS RAY DIMENSIONS AND HYDRODYNAMIC PARAMETERS

Parameters	Value
Weight	137.2 N
Buoyancy	144.6 N
Rotational Inertia	0.37, 0.52, 0.18 kgm^2
Added mass Coeff.	4.79, 12.20, 61.56 kg 6.18, 5.14, 1.44 kgm^2
Linear Drag Coeff.	0.161, 0.515, 0.003 kg/s 0.018, 0.072, 0.032 kgm^2/s
Quadratic Drag Coeff.	9.58, 12.27, 108.50 Ns^2/m^2 0.003, 0.215, 0.145 Nms^2
Centre of Buoy. (COB)	[0, 0, 0.016] m
Maximum Thrust	20 N

to estimate added-mass and damping terms, while geometry-based volume and moment-of-inertia were determined from 3D modeling tools and physical experiments. The estimated hydrodynamic coefficients employed in the drone model and controller synthesis are listed in Table I.

V. Embedded System & Software Architecture

A. Control and Communication Framework

The embedded system architecture integrates high-level and low-level control functions. A Raspberry Pi 5 serves as the companion computer to satisfy real-time video capture, processing, and streaming requirements. It runs ROS 2 as the primary middleware, while an ESP32 (micro-ROS capable) handles low-level I/O for sensors and actuators. Middleware uses the XRCE-DDS client–agent pattern so resource-constrained ESP32 nodes participate in the ROS 2 data-space via an XRCE Agent on the Pi.

The drone is tethered for power and Ethernet. The Ethernet pair runs to a surface router that relays the Pi's WebRTC video stream and DDS telemetry to the ground station, making position, sensor feeds, and state available for observation and logging. The ESP32 performs sensor data acquisition, filtering, and control-loop execution.

B. Sensor–Actuator Integration within the Embedded Architecture

Underwater operation is GPS denied; navigation relies on an onboard INS that fuses a 6-axis IMU (3-axis accelerometer and 3-axis gyroscope), a 3-axis magnetometer, and a pressure sensor for depth. IMU and magnetometer data are fused in an extended Kalman filter to reduce gyro drift and improve attitude estimates, while pressure is filtered with an exponential moving average to stabilize depth. The ESP32 maintains the state estimate and supplies it to the SMC architecture, which computes actuator commands for the six thrusters to close the loop on commanded velocities and headings (see Fig. 7).

Safety features are integrated into the embedded stack. A leak sensor inside the pressure canister triggers an immediate shutdown sequence on moisture detection. A magnetically actuated switch permits external power control without

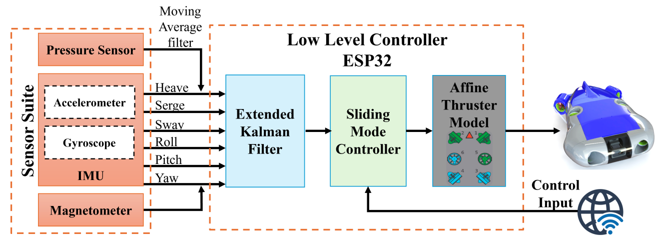


Fig. 7. Low level control architecture: Embedded control stack on ESP32 showing sensing, EKF fusion, SMC, and thruster allocation for WS Ray.

compromising watertight integrity and provides a manual intervention path during emergencies.

For actuation, six underwater thrusters (waterproof 12–24 V class, ∼2 kg static thrust, 2833-1000 KV brush-less motors) are each driven by a bidirectional ESC to enable precise reversible thrust. Camera pitch is controlled by a serial bus servo, allowing the operator to tilt the camera independent of drone attitude for improved situational awareness.

VI. Immersive and Mobile-Based Human–Machine Interfaces

A. Mouse-Form Analog Auxiliary Controller

A mouse-form auxiliary controller (see Fig. 8) was developed to enable accessible teleoperation for older adults and users with accessibility needs. This device facilitates fine, analog control while allowing the user's hand to remain in a relaxed, resting posture.The device incorporates three relatively linear push-buttons, each with 20 mm total linear travel. Each button houses an SS49E linear Hall-effect sensor while corresponding N35 NdFeB (20×3 mm) cylindrical magnets are fixed to the base; relative motion changes the magnetic field at the sensor and yields an approximately linear analog output. The first 5 mm of travel is a dead zone that produces no output; displacement from 5–20 mm is mapped linearly to a 0–1 command range. A return spring recenters each button when released, and the three analog channels are mapped to thumb, index, and middle finger inputs to provide compact proportional multi-DOF control.

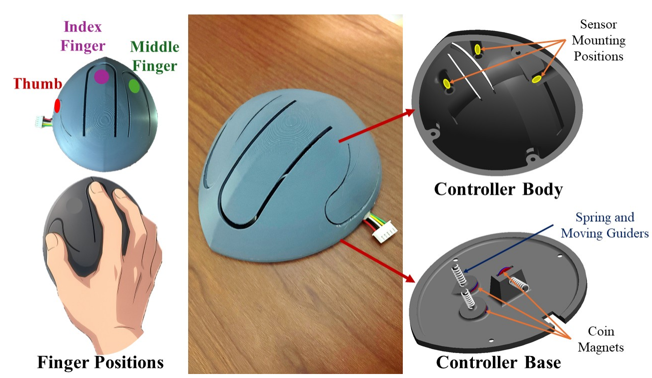


Fig. 8. Mouse-form auxiliary controller: linear Hall-effect sensors generate proportional signals based on the distance to an onboard magnet; those signals provide motion and velocity commands to the drone. The figure illustrates sensor-suite attachment, magnet placement, and a usage demonstration.

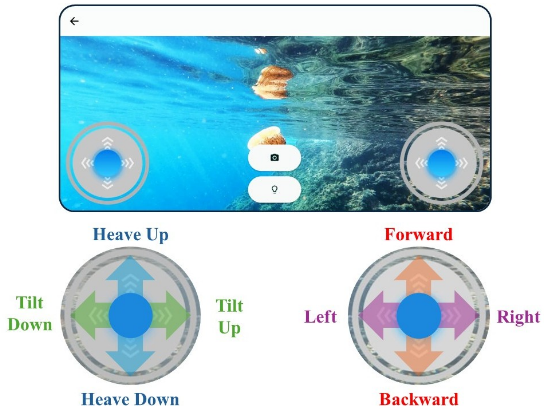


Fig. 9. Touch-based mobile teleoperation app. The interface displays the live camera feed and provides dual on-screen joysticks: the left joystick controls heave and tilt, while the right joystick controls planar translation

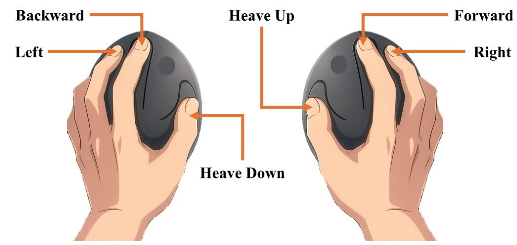


Fig. 10. Dual mouse-form auxiliary controllers operated concurrently. Each controller outputs independent analog channels mapped to finger inputs: thumbs → heave, index fingers → surge, middle fingers → sway.

Algorithm 1 Head-yaw → yaw-rate mapping

Require: planner_available $\in \{\text{true}, \text{false}\}$
Require: current head-yaw ψ_k, previous head-yaw ψ_{k-1}
Require: Parameters: ψ_{dz} (yaw dead-zone), $\Delta\psi_{\text{th}}$ (delta-yaw threshold),
Ensure: Outputs: commanded yaw rate r_{cmd} (rad/s)
Require: Control constants: k_ψ (proportional gain)

$\Delta\psi \leftarrow \psi_k - \psi_{k-1}$
$r_{\text{cmd}} \leftarrow 0$
if planner_available = **false then**
  if $|\psi_k| \leq \psi_{\text{dz}}$ **then**
    $r_{\text{cmd}} \leftarrow 0$
  else
    $r_{\text{cmd}} \leftarrow k_\psi \Delta\psi$
  end if
else
  if $(\Delta\psi > \Delta\psi_{\text{th}} \wedge \psi_k > 0) \wedge (-\Delta\psi > \Delta\psi_{\text{th}} \wedge \psi_k < 0)$ **then**
    $r_{\text{cmd}} \leftarrow k_\psi\, \psi_k$
  else
    $r_{\text{cmd}} \leftarrow 0$
  end if
end if
return r_{cmd}

B. Touch-Based Mobile Teleoperation App with Live Video Feedback

A Flutter-based mobile teleoperation application provides real-time video streaming and dual-joystick touch controls for non-VR users (see Fig. 9). Video is received using the flutter_webrtc plugin to enable real-time streaming to the phone. The app depends on http to establish initial signaling/handshake and simple Representational State Transfer (REST) interactions between the WebRTC client and the signaling/server, which is required before a peer connection can be created. Control commands are sent over a persistent web socket channel connection for low-latency, bidirectional command exchange. The on-screen joysticks are implemented with the flutter_joystick [17] package. Because the current system runs on a local network, the main page of the app allows the user to enter the Raspberry Pi IP address and connect to that host; after connecting the control page displays the live camera feed in the background and provides buttons to toggle cameras and front lights. Two on-screen joysticks provide control inputs: the right joystick maps to Planar motion while the left joystick is used for heave control and camera tilt (see Fig. 9).

C. VR-Enabled Mobile Teleoperation combined with Analog Auxiliary Controller

The VR-enabled mobile teleoperation application presents a stereoscopic-like view by splitting a single 640×480 video frame into two displayed windows. The left view renders the pixel region (0,0)–(600,480) while the right view renders (40,0)–(640,480); these overlap offsets must be carefully calibrated to avoid a doubled appearance and to provide a comfortable immersive visualization. For control, head motion is obtained via the dchs_motion_sensors dependency [18] to provide yaw, pitch, and roll inputs from the user's head.

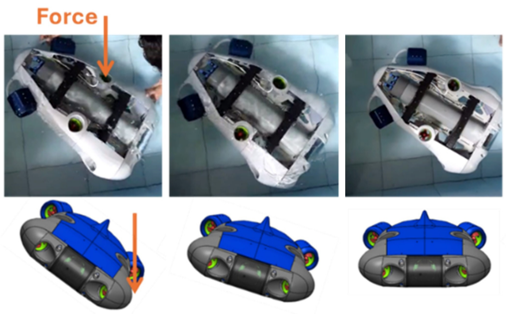
Two mouse-form auxiliary controllers are used concurrently; each controller provides independent analog channels that regulate the drone's speed and direction, mapped to the user's left and right hand finger actions (see Fig. 10). If contradictory actions occur simultaneously (e.g., both heave-up and heave-down pressed) the system issues a stop command to the drone. Linear-motion control combined with head yaw is handled in two distinct phases: (1) turning while the drone is in linear motion, and (2) turning in-place (stationary yaw adjustment). These cases execute different control logic as outlined in Algorithm 1. Camera tilt is mapped directly to head-pitch measurements and is not combined with inputs from the auxiliary controllers.

VII. Results & Discussion

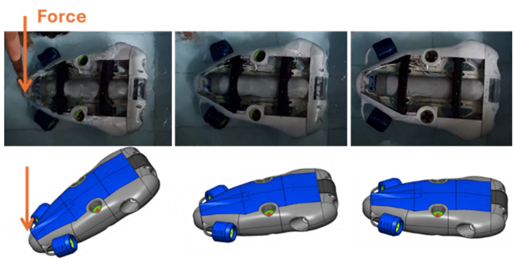
WS Ray was tested to verify the functionalities, and the following sections include each result in detail.

A. Hydrostatic performance : pitch and roll stability

Passive roll stabilization was verified in calm-water trials by placing the drone afloat without active control and



(a)



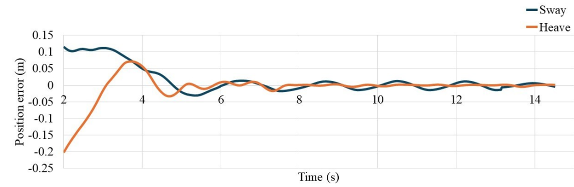
(b)

Fig. 11. Hydrostatic stability of WS Ray. (a) Passive roll stabilization performance. (b) Passive pitch stabilization performance, demonstrating that the COB and COG lie in the same plane, with the COB located above the COG.

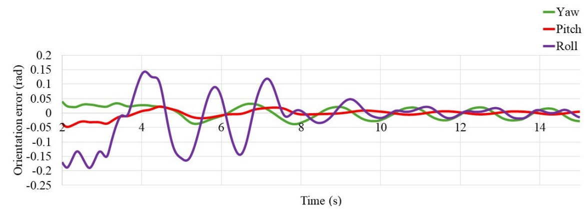
observing its free response (see Fig. 11a); the drone maintained a near-horizontal orientation, confirming inherent roll stability. Longitudinal trim and pitch stability were calibrated using the internal weight-balancing mechanism: coarse trim adjustments were made by adding or removing 3 mm steel plates, and fine pitch trim was achieved by sliding the weight stack along the central axis of the mounting plate. Following adjustment the drone held a neutral pitch in water and returned to level after external disturbances (see Fig. 11b). These tests confirm the design provides hydrostatic stability and permits straightforward reconfiguration of mass distribution without requiring active compensation.

B. Dynamic motion response : surge, sway, and heave

Tests were performed in a 2 m-deep pool to evaluate translational response in surge, sway and heave. Peak velocities were: forward (surge) $0.50\,\mathrm{m{\cdot}s^{-1}}$, lateral (sway) $0.30\,\mathrm{m{\cdot}s^{-1}}$, heave-up $0.20\,\mathrm{m{\cdot}s^{-1}}$, and heave-down $0.10\,\mathrm{m{\cdot}s^{-1}}$ (all reached in 0.5 s), reflecting the drone's slight positive buoyancy and resulting asymmetric vertical response. The performance of the SMC under sudden disturbances was evaluated by applying a lateral displacement to the drone during forward motion. The controller successfully stabilized the drone and enabled it to recover its commanded trajectory (see Fig. 12); gains used were $K = [5, 5, 5, 0.5, 1, 2]$ and $\lambda = [5, 5, 5, 1, 1, 1]$ for [surge, sway, heave, roll, pitch, yaw]. These motion tests demonstrate the drone's commanded-motion performance and control authority, and confirm operation to the tested 2 m immersion depth.



(a)



(b)

Fig. 12. Trajectory error recovery over time following a sudden lateral displacement applied during forward motion at 0.50 $\mathrm{m{\cdot}s^{-1}}$. (a) Sway and heave recovery. (b) Yaw, pitch, and roll recovery.

C. HMI performance : video throughput and control responsiveness

Video feed operated at 28 fps, consistent with a real-time stream. Stereo VR calibration produced no double-image artifacts and head-tracking delivered the expected motion. Mouse-form controller and on-screen joystick inputs provided acceptable teleoperation responsiveness during trials. Overall, the HMI and real-time performance met the operational requirements.

VIII. Conclusions

This paper presented the development of an underwater drone for Internet-based teleoperation. The mechanical design and control algorithm performed as expected, and a mobile app together with a VR-based HMI provide an immersive, easy-to-use interface for remote users. Planned future work includes full remote Internet control (currently implemented on localhost) and development of a collaborative tether-management system to deploy and retrieve the drone; additionally, improvement of the proposed auxiliary controller is identified as a future direction because it may be beneficial across a range of applications. In conclusion, the successful demonstration of the WS Ray system not only validates the proposed design and control architecture but also establishes a robust and low-cost framework that can pave the way for broader accessibility to underwater exploration and research.

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